BEST CASE STUDY COMPETITION 2026 | THE LHOFT | TARGET: STOKR MIFID II SECURITY EX-MICAR ART. 2(4)(A)

Tokenised Solar Debt Instrument: A €2.0M Institutional Framework

2.5 MWp Commercial & Industrial (C&I) Solar Installation in Luxembourg • Segregated SV • €500 Retail Access

1. Executive Summary & Verifiable European Precedents

PROBLEM & VALUE PROPOSITION

This proposal outlines the tokenisation of a **€2.0M senior secured debt instrument** financing a 2.5 MWp Commercial & Industrial (C&I) solar photovoltaic installation in Luxembourg. Structured strictly as a MiFID II compliant "Transferable Security," the instrument resolves the duration mismatch between 30-year solar infrastructure assets and 10-year capital market liquidity horizons. By integrating IoT smart meter telemetry with an institutional smart contract framework, the solution lowers minimum retail investment tickets from €100,000+ to **€500** (4,000 tokens) while cutting 10-year servicing overhead by 68.2% (€167,000 saved).

Dual Offtake Pricing & Real-World Precedents: In Luxembourg, a 2.5 MWp (>500 kWp) solar plant monetises power via two routes: **(1) Grid Injection / Tender Floor (€70–€86/MWh)** into the public grid (ILR wholesale benchmark €86/MWh), or **(2) On-Site C&I Corporate PPA (€90–€125/MWh)** supplying a corporate facility behind-the-meter (saving the buyer Creos network tariffs vs. €164+/MWh retail rates). We anchor our baseline on the conservative **€75/MWh grid PPA floor** (yielding a **1.34x DSCR**), treating €95–€125/MWh corporate PPAs as upside (**1.80x–2.48x DSCR**). Our design builds on verifiable European precedents: **STOKR's Luxembourg SV notes** (*BMN1/BMN2, STRC Note*) and **STOKR's Private Credit Registry**; Germany's BaFin-regulated **€3.0M Ratekau Solar Park tokenised bond** (Cashlink/FINEXITY); and **BNP Paribas & EDF ENR's** digital solar bond.

2. AI & Digital Approach: Smart Contract & Oracle Architecture

HYBRID ON/OFF-CHAIN FLOW

Multi-Chain Institutional Architecture: The €2.0M solar bond is prototyped on an **upgradeable EVM smart contract** (`UUPS` proxy) designed for seamless cross-chain portability to STOKR's flagship **Bitcoin Liquid Network (Blockstream AMP)** architecture. Rather than tracking speculative crypto prices, a decentralized Chainlink Oracle network (`AggregatorV3Interface`) ingests daily generation data (in MWh) from physical IoT smart meters, anchoring coupon distributions to contracted PPA cash flows. Following the Bank for International Settlements (BIS) Project Genesis standard to prevent "immutable garbage-in" sensor risk, on-chain IoT meter feeds undergo mandatory **quarterly off-chain reconciliation** against official grid settlement statements from **Creos Luxembourg S.A.** (Luxembourg's national electricity grid operator) and the **ILR** (national regulator) prior to coupon release.

AI Predictive Maintenance & Human-in-the-Loop: While coupon accounting follows deterministic smart contract rules, an off-chain unsupervised machine learning model (**Isolation Forest**) monitors real-time IoT plane-of-array irradiance and string inverter Performance Ratios (PR). It flags multi-dimensional hardware degradation weeks before Debt Service Coverage Ratio (DSCR) erosion occurs. STOKR compliance officers retain exclusive **human-in-the-loop authority** over on-chain KYC whitelists, emergency circuit breakers, and dispatching Operations & Maintenance (O&M) crews upon AI alerts.

Figure 1.1: End-to-End Institutional Issuance, Telemetry Verification & Distribution Architecture

`UUPS Proxy / Liquid AMP Compatible`



3. Fairness, Error Correction & Bias Reduction

ACCESSIBILITY & SAFEGUARDS

Financial Inclusion & Cognitive Fairness: Conventional renewable infrastructure debt inherently excludes retail citizens due to €100,000+ minimum private placement thresholds. Tokenisation democratises access to **€500 per token**. To ensure cognitive fairness and prevent technical bias, all UUPS smart contract calls, Blockstream AMP asset rules, and Chainlink Oracle feeds are abstracted behind **STOKR's intuitive Web2 investor dashboard**, enabling seamless SEPA/Euro subscriptions without requiring self-custody cryptography expertise.

Institutional Error Correction (Zero Permanent Loss): A critical risk of permissionless blockchains is irreversible asset loss from stolen keys or wallet hacks. Our architecture eliminates this flaw through permissioned governance. Upon verified identity checks or a court order, STOKR's `COMPLIANCE_ROLE` can atomically **freeze a compromised address, wipe the illicitly held tokens, and reissue replacement tokens** to the investor's verified backup wallet. Simultaneously, a `PAUSER_ROLE` acts as an emergency circuit breaker, halting secondary transfers during severe Oracle discrepancies or regulatory inquiries.

PLAIN-LANGUAGE INVESTOR FACTSHEET (INFORMATION NOTE SUMMARY)

RETAIL DISCLOSURE

- **The Asset & What You Own:** You hold regulated digital debt tokens (€500 per token, €2.0M total issue across 4,000 tokens) financing a 2.5 MWp C&I solar installation in Luxembourg. You own a **direct senior claim on contracted electricity revenues**—not the physical solar panels.
- **Your Returns & Offtake Pricing (5.5% Bullet Bond):** You receive a fixed **5.50% annual coupon** (€27.50/yr per €500 token) over 10 years. Even under a conservative €75/MWh grid-injection PPA floor, cash flows cover interest **1.34x over**; if power is sold directly to an industrial buyer via an on-site C&I PPA (€95–€125/MWh), coverage rises to **1.80x–2.48x**.
- **Principal Repayment at Year 10:** This is an interest-only bullet bond. Your €500 principal per token is repaid in full at Year 10 via a pre-programmed refinancing event or secondary buyout backed by the plant's remaining **20 years of productive cash flows** on its 30-year lifespan (**residual Year-10 NPV of €1.63M + €375k reserve buffer = €2.00M+ total coverage**).
- **Downside Resilience & Limited Recourse:** Under a severe 1-in-10-year P90 weather shock (−20% sunlight to 2,000 MWh), project cash flow (€110,000) still covers 100% of your interest (**1.00x Breakeven DSCR**). Issued via a segregated **Luxembourg Securitisation Vehicle (SV)**; STOKR does not guarantee repayment, and default claims are enforced exclusively against the SV's solar hardware and locked PPA receivables.
- **Liquidity & Access Notice:** Tokens are transferable only among KYC-whitelisted investors on STOKR's bulletin board. **Tokenisation does not guarantee secondary market liquidity.**

4. Implementation Plan (12-Month Roadmap & Pilot)

EXECUTION & GOVERNANCE

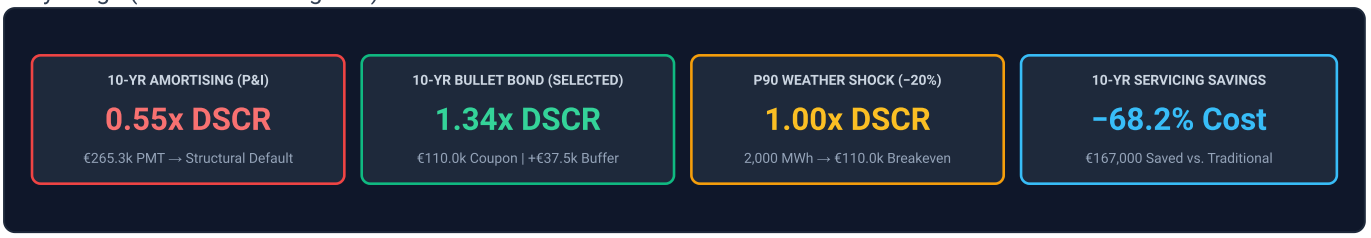
PHASE & TIMELINE	RESPONSIBLE LEADS	CORE DELIVERABLES & PILOT SCOPE	PRE-DEPLOYMENT GATE & COST
`Phase 1: M1–M3` **Legal & SV Setup**	Luxembourg Legal Counsel & Structurers	Incorporate Unregulated Securitisation Vehicle (SV); draft Information Note (<€8.0M exemption); register 10-yr PPA liens in **STOKR Private Credit Registry**.	Legal Opinion on MiFID II Security status. *Est. Cost: €25,000*
`Phase 2: M4–M6` **Pilot & Audit**	Lead Solidity/AMP Dev & External Auditors	Small-scale testnet pilot (Sepolia & Liquid AMP); test `AggregatorV3Interface` IoT meter sync, Isolation Forest AI alerts, and Freeze/Wipe/Reissue hooks.	Zero-critical-bug smart contract audit report. *Est. Cost: €15,000*
`Phase 3: M7–M9` **Portal & KYC**	STOKR Integration & Compliance Team	Connect audited contract to STOKR Web2 portal; automate off-chain KYC to on-chain whitelist; establish **Creos/ILR** quarterly grid reconciliation workflow.	End-to-end UAT & CSSF notification check. *Est. Cost: €8,000*
`Phase 4: M10–M12` **Live Issuance**	Issuer SV Board & Paying Agent	Launch €2.0M subscription (4,000 tokens @ €500); lock initial DSRA reserve buffer; activate live AI predictive maintenance & quarterly investor reporting.	€2.0M escrow close & Oracle heartbeat live. *Servicing: €3k/yr*

5. Impact, Financial Viability & Structural Pivot

QUANTITATIVE PROOF

Resolving the Infrastructure Duration Mismatch: Modeled on authoritative European benchmarks (P50 yield of 2,500 MWh × conservative €75/MWh Luxembourg grid PPA floor − €40,000 fixed OpEx), the solar installation generates an **Annual Operating Cash Flow (OCF) of €147,500**. Stress-testing a conventional 10-year fully amortising loan exposed a fatal structural flaw: constant principal and interest payments require **€265,346/yr**, causing an immediate default at **0.55x DSCR**. Forcing a 30-year solar asset to amortise its full CapEx in 10 years destroys project viability. Pivoting to a **10-Year Bullet Bond (Interest-Only at 5.5%)** reduces annual debt service to €110,000/yr, achieving a robust **1.34x Baseline DSCR** (+€37,500/yr cash buffer), maintaining a **1.00x Breakeven DSCR** under a −20% P90 weather shock (2,000 MWh), and reaching **1.80x DSCR** if contracted via an on-site C&I Corporate PPA (€95/MWh). At Year 10, discounting the remaining 20 years of operation (Years 11–30) at 6.50% WACC yields a residual NPV of **€1,625,230**, which combined with the **€375,000** reserve buffer provides **€2,000,230 (100.01% principal coverage)**.

Three Measurable Impact Indicators (Assumptions Benchmarked in Appendix): (1) Retail Inclusion: 99.5% reduction in minimum entry ticket (€100,000 to €500), assessed via whitelisted retail participation rate; **(2) Lifecycle Cost Compression: 68.2% reduction** in 10-year issuance and paying-agent fees (€78,000 tokenised vs. €245,000 conventional private placement, saving €167,000); and **(3) AI Cash Flow Protection:** Preserving a **1.32x DSCR** during string inverter faults via Isolation Forest early triage (vs. 1.14x unmitigated).



6. Risks, Legal Classification & Specialist Approvals

MIFID II & SV BOUNDARIES

Regulatory Classification & Approvals: Because the tokens represent transferable senior debt claims, they qualify strictly as **"Transferable Securities" under MiFID II (Article 4(1)(44))** and are explicitly excluded from the EU Markets in Crypto-Assets Regulation under **MiCAR Article 2(4)(a)**. Under the **Luxembourg Prospectus Law of 16 July 2019**, public offerings below €8.0M are exempt from a full CSSF-approved prospectus, requiring instead a standardised **Information Note** filed with the CSSF alongside specialist legal opinions, smart contract security audits, and AML/KYC onboarding checks.

Limited Recourse, Double-Pledging & Data Integrity Risks: The bond is issued via an **Unregulated Securitisation Vehicle (SV)** under the Luxembourg Securitisation Law of 22 March 2004 (as amended), establishing a strict **Limited Recourse** boundary. STOKR S.A. acts solely as technology provider and does not guarantee repayment. To mitigate credit and operational risks, PPA receivables are locked in **STOKR's Private Credit Registry** to prevent double-pledging, while quarterly **Creos Luxembourg S.A.** grid settlement audits eliminate Oracle manipulation risk.

7. Tools, Assumptions & Methodological Disclosures

TRANSPARENCY & CITATIONS

- **Interactive Financial & Compliance Portal:** Custom HTML/JavaScript/Python stress-testing simulator and Web2 compliance console developed to model DSCR sensitivities, Isolation Forest PR anomalies, and UUPS governance hooks.
- **Smart Contract & Multi-Chain Prototyping:** Conceptualized using Foundry (OpenZeppelin `UUPSUpgradeable`, Chainlink `AggregatorV3Interface`) with compliance state-machine mapping to Bitcoin Liquid Network (Blockstream AMP).
- **Generative AI Disclosure:** Google Gemini was utilized to refine technical English syntax, structure plain-language retail disclosures, and cross-verify European regulatory citations; all financial calculations were independently verified by the author.
- **Modeled Assumptions & Fictional Data:** Per competition rules, the issuer entity is fictional. All parameters (2.5 MWp in Luxembourg, €2.0M CapEx, €40k OpEx, 2,500 MWh P50 yield, €75/MWh grid PPA floor vs. €95–€125/MWh C&I PPA, 5.5% coupon, 30-year solar lifespan) are benchmarked to Fraunhofer ISE, EU JRC PVGIS, ILR, Creos, and Pexapark (see 6-page Appendix).